

Lecture 7. Optimization: Gradient descent and constrained optimization.

→ X_0 → Loss function $\mathcal{L}(\tilde{y}, \hat{y})$ to be minimized.

→ X_0 → $\frac{\partial \mathcal{L}(\tilde{y}, \hat{y})}{\partial W^{(e)}}$, $\frac{\partial \mathcal{L}(\tilde{y}, \hat{y})}{\partial \tilde{b}^{(e)}} \Rightarrow$ searching direction

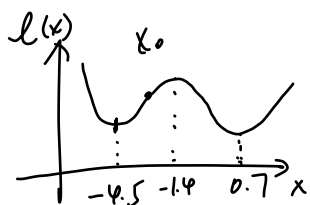
$\Rightarrow$ numerical optimization technique for searching strategy.

Remark: ① assuming that the objective function (loss function) is differentiable.

② mainly focus on "minimization" objectives.

Example: consider the loss function $\mathcal{L}(x) = x^4 + 7x^3 + 5x^2 - 17x + 3$ ✓

The gradient: $\frac{d\mathcal{L}(x)}{dx} = 4x^3 + 21x^2 + 10x - 17$ $\frac{d^2\mathcal{L}(x)}{dx^2} = 12x^2 + 42x + 10$.



$\frac{d\mathcal{L}(x)}{dx} = 0$ $x_1 = -4.5$ $\frac{d^2\mathcal{L}(x)}{dx^2} = 64$
 $x_2 = -1.4$ $\frac{d^2\mathcal{L}(x)}{dx^2} = -25.282$ (1-st derivative, decreasing)
 $x_3 = 0.7$ $\frac{d^2\mathcal{L}(x)}{dx^2} = 299.88$ (local maximum)

$\Rightarrow$ start at some x_0 , negative gradient, leads us to some (local) minimum

$\Rightarrow$ Hence: minimization of the objective $\Rightarrow$ gradient descent
 maximization of the objective $\Rightarrow$ gradient ascent

1. Unconstrained optimization

Problem: Solving the minimum of a real-valued function.

$$\min_{\vec{x} \in \mathbb{R}^d} f(\vec{x})$$

where $f: \mathbb{R}^d \mapsto \mathbb{R}$ being a differentiable objective function

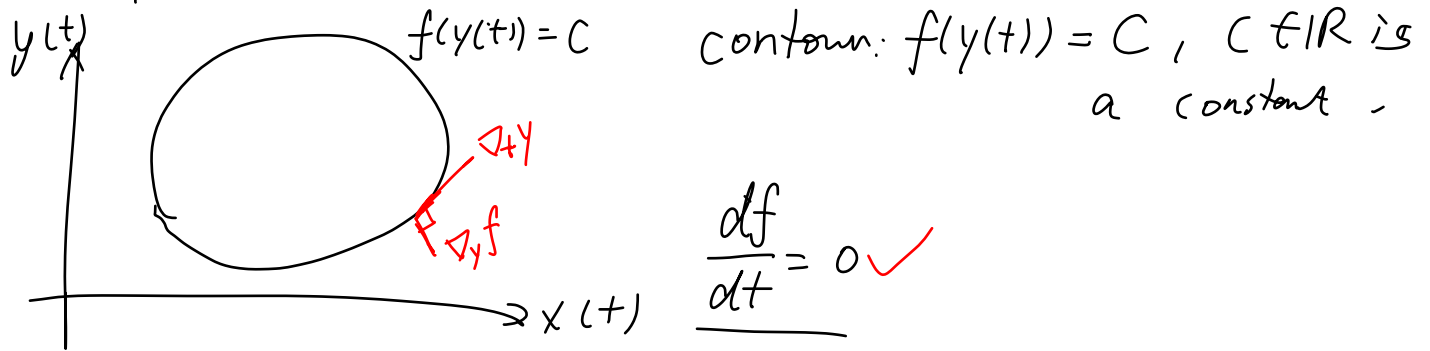
Solution: starting from a particular location x_0 , searching is iteratively done by $\gamma_t > 0$, step-size (learning rate).

$$\star \vec{x}_{t+1} = \vec{x}_t - \gamma_t (\nabla f(\vec{x}_t))^T$$

Goal: $f(\vec{x}_0) \geq f(\vec{x}_1) \geq \dots$ converge to a local minimum.

Remark: The gradient points in a direction orthogonal to the contour lines of the function

Let: $\vec{y}(t) = [x(t) \ y(t)] \in \mathbb{R}^2$ being curve.



Meanwhile:
$$\frac{df}{dt} = \frac{df}{dy} \cdot \frac{dy}{dt} = \left[\frac{\partial f}{\partial x} \ \frac{\partial f}{\partial y} \right] \begin{bmatrix} \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial t} \end{bmatrix} = \langle \nabla_x f, \nabla_y \rangle = 0$$

2. Configuration of step-size.

General idea: adaptive gradient decent instead of fixed-size.

$\Rightarrow$ rescale the step-size at each iteration:

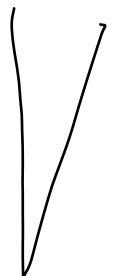
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- ① Function value $\uparrow$ after a gradient step
 $\rightarrow$ undo, step-sized
 - ② function value $\downarrow$ after a gradient step
 $\rightarrow$ try to increase the step-size

3. Gradient descent with momentum.

Motivation: The convergence of gradient decent could be slow due to the curvature of the optimization surface.

Example: valley-shape surface

$\rightarrow$ curved in one direction
flat in another.



Idea: Extra memory for the gradient decent
 consider $\alpha \in [0, 1]$.

$$\Delta X_t = X_t - X_{t-1}$$

$$\vec{X}_{t+1} = \vec{X}_t - \gamma_t ((\nabla f)(X_t))^T + \alpha \Delta X_t$$

$$\begin{aligned} \vec{X}_{t+1} - \vec{X}_t &= \underline{\Delta \vec{X}_{t+1}} = -\gamma_t ((\nabla f)(\vec{X}_t))^T + \alpha \Delta \vec{X}_t \\ &= -\sum_{\tau=1}^t \underbrace{\alpha^{t-\tau} \gamma_\tau}_{\gamma'_\tau} ((\nabla f)(\vec{X}_\tau))^T \end{aligned}$$

i.e. update the new value according to "accummulated gradient with dynamical rate".

Case 1: direction of recent gradients are consistent.
 $|\Delta X_k| \uparrow$ (accelerating the iteration at the beginning)

Case 2: direction of recent gradient diverse: $|\Delta X_k| \downarrow$
 (stablizing the iteration at the late stage).

4. Stochastic Gradient Decent

Consider the objective function

$L(\theta) = \sum_{i=1}^N L_i(\theta)$, with is the sum of losses L_i incurred by each sample i , θ is the parameter to be optimized.

Target: Find θ that minimized L .

$\Rightarrow$ consider data set $D\{(x, y)\}^n$ real value

$$\theta_{t+1} = \theta_t - \gamma_t \sum_{i=1}^N \frac{\partial L(y^{(n)}, f(\vec{x}^{(n)}, \theta))}{\partial \theta}$$

predicted value for the given θ .

Problem: Difficulties for computing the gradient, when.

① Training data set is enormous

② evaluating the sum of gradient is not easy

Solution: use empirical loss to approximate expected loss

an unbiased estimate of the true gradient by randomly select a small, mini-batch of data

Benefit of mini-batch

- ① quick to estimate
- ② noisy estimation allows us to get out of some bad local optima
- ③ good for generalization.

Exercise: Consider $f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \frac{1}{2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} 2 & 1 \\ 1 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

run gradient decent and starting at $\vec{x}_0 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$, what is $\vec{x}_1$ considering $\gamma = 0.085$

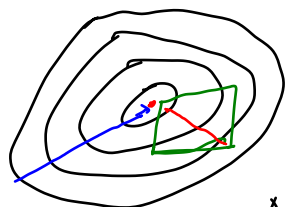
Answer: Let $\vec{x} = [x_1, x_2]^T$, ① $f(\vec{x}) = \vec{x}^T \underline{A} \vec{x} - \underline{B} \vec{x}$, $\frac{df}{d\vec{x}} = \vec{x}^T \underline{A} - \underline{B}$

$$\textcircled{2} \vec{x}_0 = [-3 \ -1]^T, \nabla f(\vec{x}_0) = [-3 \ -1]^T \begin{bmatrix} 2 & 1 \\ 1 & 20 \end{bmatrix} - \begin{bmatrix} 5 & 3 \end{bmatrix} = [-12 \ -26]$$

$$\vec{x}_1 = \vec{x}_0 - \gamma (\nabla f(\vec{x}_0))^T = \begin{bmatrix} -3 \\ -1 \end{bmatrix} - 0.085 \begin{bmatrix} -12 \\ -26 \end{bmatrix} = \begin{bmatrix} -1.98 \\ 1.21 \end{bmatrix}$$

5. Constrained Optimization.

consider minimizing the objective function.



$$\begin{array}{ll} \min_{\vec{x}} f(\vec{x}) & \text{primal problem} \\ \text{subject to } g_i(\vec{x}) \leq 0, & \text{for all } i=1, \dots, m \end{array}$$

An easy (but not implementation-friendly) unconstrained objective.

$$\min_{\vec{x} \in \mathbb{R}^n} J(\vec{x}) := f(\vec{x}) + \sum_{i=1}^m I(g_i(\vec{x})) \quad \text{in which } I(z) = \begin{cases} 0, & \text{if } z \leq 0 \\ \infty & \text{otherwise} \end{cases}$$

$I(z)$ is an infinite step function, i.e. $I(z) =$

only solution with $z \leq 0$ would be found, but difficult to implement.

Instead, an alternative approach: Lagrange Multiplier

For $\lambda_i \geq 0$, define.

$$\mathcal{L}(\vec{x}, \vec{\lambda}) := f(\vec{x}) + \sum_{i=1}^m \lambda_i g_i(\vec{x})$$

$$\checkmark = \underline{f(\vec{x}) + \vec{\lambda}^T \vec{g}(\vec{x})}, \quad \vec{\lambda} = [\lambda_1, \dots, \lambda_m]^T$$

$$\vec{g}(\vec{x}) = [g_1(\vec{x}), \dots, g_m(\vec{x})]^T$$

Then, we can propose the corresponding **dual problem**

dual problem $\rightarrow$

$$\begin{aligned} & \max_{\vec{\lambda} \in \mathbb{R}^m} D(\vec{\lambda}) \\ & \text{subject to } \vec{\lambda} \geq \vec{0}, \text{ with } D(\vec{\lambda}) := \min_{x \in \mathbb{R}^d} \mathcal{L}(x, \vec{\lambda}) \end{aligned}$$

x : primal variable; λ : dual variable

$\vec{\lambda} \geq \vec{0} \Leftrightarrow \lambda_i \geq 0$ for all $i \in [1, m]$ unconstrained optimization problem when λ is fixed.

Remark:

- ① $D(\lambda)$ is easy to solve when λ is fixed.
- ② $\mathcal{L}(\vec{x}, \vec{\lambda})$ is affine w.r.t. $\vec{\lambda}$, hence, $D(\vec{\lambda}) := \min_{x \in \mathbb{R}^d} \mathcal{L}(\vec{x}, \vec{\lambda})$ is a point-wise minimum of affine function of λ .
- ③ The outer problem over λ can be solved efficiently ($D(\lambda)$ is concave)

We need Minmax inequality.

For any function φ with two argument x, y .

$$\max_y \min_x \varphi(x, y) \leq \min_x \max_y \varphi(x, y)$$

Proof: For any x_0, y_0 , one can verify.

$$\min_x \varphi(x, y_0) \leq \varphi(x_0, y_0) \leq \max_y \varphi(x_0, y) \quad \textcircled{1} \checkmark$$

Consider $\underline{y_0} := \arg \max_y \boxed{\min_x \varphi(x, y)}$ $\underline{x_0} := \arg \min_x \boxed{\max_y \varphi(x, y)}$

since ① holds for any arbitrary x_0, y_0 $\uparrow$

$$\min_x \varphi(x, \underline{y_0}) \leq \max_y \varphi(\underline{x_0}, y)$$

$$\Rightarrow \max_y \min_x \varphi(x, y) \leq \min_x \max_y \varphi(x, y) \quad \text{Q.E.D.}$$

From minmax inequality -

$$\min_{\vec{x} \in \mathbb{R}^m} \max_{\vec{\lambda} \geq \vec{0}} \mathcal{L}(\vec{x}, \vec{\lambda}) \geq \boxed{\max_{\vec{\lambda} \geq \vec{0}} \min_{\vec{x} \in \mathbb{R}^m} \mathcal{L}(\vec{x}, \vec{\lambda})} \quad \text{dual problem}$$

note that if $\vec{\lambda} \geq \vec{0}$, we relaxed the indication function to a linear function. (linear combination among $g_i(\vec{x})$, $i \in [1, m]$)

$$\mathcal{L}(\vec{x}, \vec{\lambda}) \leq J(\vec{x}) (= f(\vec{x}) + \sum_{i=1}^m \lambda_i g_i(\vec{x}))$$

$$\Rightarrow J(\vec{x}) = \max_{\vec{\lambda} \geq \vec{0}} \mathcal{L}(\vec{x}, \vec{\lambda}) \quad \text{consider that the original problem}$$

$$\text{aim at } \min_{x \in \mathbb{R}^d} J(x) \Leftrightarrow \min_{x \in \mathbb{R}^d} \max_{\vec{\lambda} \geq \vec{0}} \mathcal{L}(\vec{x}, \vec{\lambda})$$

$$\Rightarrow \max_{\vec{\lambda} \geq \vec{0}} \min_{x \in \mathbb{R}^d} \mathcal{L}(x, \vec{\lambda}) = \max_{\vec{\lambda} \in \mathbb{R}^m} D(\vec{\lambda}) \leq \min_{x \in \mathbb{R}^d} \max_{\vec{\lambda} \geq \vec{0}} \mathcal{L}(\vec{x}, \vec{\lambda})$$

$\Rightarrow$ optimal value of the objective function for the dual problems is a lower bound of that of the original problem.

Remark: In case equality constraints

$$h_j(\vec{x}) = 0 \quad \begin{cases} h_j(\vec{x}) \leq 0 \\ -h_j(\vec{x}) \leq 0 \end{cases}$$

$\nwarrow$ weak duality

In particular:

$$\text{we define optimal duality gap} = \min_{x \in \mathbb{R}^m} \max_{\vec{\lambda} \geq \vec{0}} \mathcal{L}(\vec{x}, \vec{\lambda}) - \max_{\vec{\lambda} \geq \vec{0}} \min_{x \in \mathbb{R}^m} \mathcal{L}(\vec{x}, \vec{\lambda})$$

Additionally, optimal duality gap = 0 (strong duality hold)

① $g_i(x)$, $i \in [1, m]$ are convex

② $\exists x$, s.t. $g_i(x) < 0 \quad \forall i \in [1, m]$ (Slater's condition)

6. convex programming / optimization

1.1 Def: Convex Optimization is a class of optimization problems

$$\begin{aligned} \min_x & f(x) \\ \text{s.t.} & g_i(x) \leq 0, \quad \forall i \in [1, m] \end{aligned}$$

with $f(\cdot)$ is a convex function, $g_i(x) \leq 0$, $\forall i \in [1, m]$ form convex set.

(2) convex set: A set C is convex if for any $\vec{x}, \vec{y} \in C$, one has $\forall \alpha \in [0, 1]$, $\alpha \vec{x} + (1-\alpha) \vec{y} \in C$



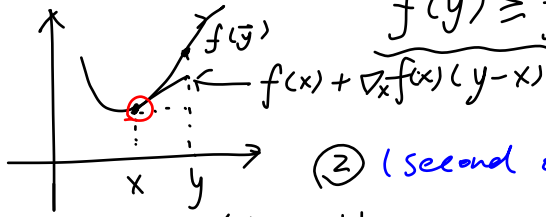
(3) convex function: A function $f: C \subseteq \mathbb{R}^D \rightarrow \mathbb{R}$ is convex if for any $\vec{x}, \vec{y} \in C$

$$\forall \alpha \in [0, 1], f((1-\alpha)\vec{x} + \alpha\vec{y}) \leq (1-\alpha)f(\vec{x}) + \alpha f(\vec{y}) \quad \checkmark$$

① Equivalently, if f is differentiable (i.e. $\nabla f(x)$ exists for all $\vec{x} \in C$)

Then f is convex if and only if for all $\vec{x}, \vec{y} \in C$

$$f(\vec{y}) \geq f(\vec{x}) + \nabla_x f(x)^T (y - x) \quad (\text{first order condition})$$



② (second order condition) If $f(x)$ is twice differentiable (i.e. Hessian matrix exists for all $\vec{x} \in C$)

$$f: \mathbb{R}^m \rightarrow \mathbb{R}$$

$$\nabla^2 f(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$$

$$i=1, \dots, m, j=1, \dots, m$$

$f(x)$ is convex $\iff \nabla_x^2 f(x)$ is positive semidefinite

Remark ① for convex loss functions in unconstrained optimization, all local minimum are global minimum.

② a concave function is the negative of a convex function.

Theorem. Given nonnegative reals α_1 and true convex functions f_1, f_2 then $\alpha f_1 + (1-\alpha)f_2$ is still convex, $\alpha \in [0, 1]$.

Proof: By definition, $\forall \vec{x}, \vec{y} \in C$

$$f_1(\alpha \vec{x} + (1-\alpha)\vec{y}) \leq \alpha f_1(\vec{x}) + (1-\alpha)f_1(\vec{y}) \quad (1)$$

$$f_2(\alpha \vec{x} + (1-\alpha)\vec{y}) \leq \alpha f_2(\vec{x}) + (1-\alpha)f_2(\vec{y}) \quad (2)$$

$$f := f_1 + f_2.$$

$$f(\alpha \vec{x} + (1-\alpha)\vec{y})$$

$$= f_1(\alpha \vec{x} + (1-\alpha)\vec{y}) + f_2(\alpha \vec{x} + (1-\alpha)\vec{y})$$

$$\begin{aligned}
&\leq \alpha f_1(\bar{x}) + (1-\alpha)f_1(\bar{y}) + \alpha f_2(\bar{x}) + (1-\alpha)f_2(\bar{y}) \quad (\text{since } ① \text{ } ②) \\
&= \alpha (\underline{f_1(\bar{x}) + f_2(\bar{x})}) + (1-\alpha) (\underline{f_1(\bar{y}) + f_2(\bar{y})}) \\
&= \alpha f(\bar{x}) + (1-\alpha)f(\bar{y}) \\
&\Rightarrow f(\alpha \bar{x} + (1-\alpha)\bar{y}) \leq \alpha f(\bar{x}) + (1-\alpha)f(\bar{y}) \Rightarrow f_1 + f_2 \text{ is convex } ③
\end{aligned}$$

since for any $\gamma \geq 0$ if f is convex, then

$$\gamma f((1-\alpha)\bar{x} + \alpha\bar{y}) \leq (1-\alpha)\gamma f(\bar{x}) + \alpha\gamma f(\bar{y}) \Rightarrow \gamma f \text{ is convex } ④$$

combining ③ & ④, we have for any $\alpha, \beta \geq 0$ if f_1 & f_2 are convex, $\alpha f_1 + \beta f_2$ is convex.

★ Remark: natural extension: nonnegative weight sum of more than two convex functions are still convex.

7.1 Linear programming: (special case of convex programming, with all functions are linear)

$$\begin{aligned}
&\min_{\bar{x} \in \mathbb{R}^m} \bar{c}^T \bar{x} \\
&\text{subject to } A\bar{x} \leq \bar{b}, \text{ with } A \in \mathbb{R}^{d \times m}, \bar{b} \in \mathbb{R}^d, \bar{c} \in \mathbb{R}^m
\end{aligned}$$

The Lagrangian.

$$\begin{aligned}
\mathcal{L}(\bar{x}, \bar{\lambda}) &= \bar{c}^T \bar{x} + \bar{\lambda}^T (A\bar{x} - \bar{b}) \\
&\text{with } \bar{\lambda} \in \mathbb{R}^d \text{ is vector of non-negative Lagrange multipliers}
\end{aligned}$$

$$\mathcal{L}(\bar{x}, \bar{\lambda}) = (\bar{c} + A^T \bar{\lambda})^T \bar{x} - \bar{\lambda}^T \bar{b}$$

$$\checkmark \quad \frac{\partial \mathcal{L}(\bar{x}, \bar{\lambda})}{\partial \bar{x}} = \bar{c} + A^T \bar{\lambda} \quad \text{since for } \min \mathcal{L}(\bar{x}, \bar{\lambda}), \quad \frac{\partial \mathcal{L}(\bar{x}, \bar{\lambda})}{\partial \bar{x}} = 0$$

$$\Rightarrow \text{the dual Lagrangian. } D(\bar{\lambda}) = \underline{-\bar{\lambda}^T \bar{b}}$$

Therefore, the dual optimization problem is

$$\begin{aligned}
&\max_{\bar{\lambda} \in \mathbb{R}^d} -\bar{b}^T \bar{\lambda} \\
&\text{subject to } \bar{c} + A^T \bar{\lambda} = 0, \bar{\lambda} \geq 0
\end{aligned}$$

Remark: ① since linear programming is convex programming dual problem has the same solution as the primal one.

② solving the primal problem or dual program depending on whether m or d is larger.

8. Quadratic programming

Consider the case of a convex quadratic objective function, where the constraints are affine, i.e.,

$$\min_{\vec{x} \in \mathbb{R}^m} \frac{1}{2} \vec{x}^T Q \vec{x} + \vec{c}^T \vec{x} \quad \leftarrow \text{primal}$$

subject to $A\vec{x} \leq \vec{b}$, where

① $A \in \mathbb{R}^{d \times m}$, $\vec{b} \in \mathbb{R}^d$ and $\vec{c} \in \mathbb{R}^m$

② $Q \in \mathbb{R}^{m \times m}$ being positive definite.

(m variables and d linear constraints)

The Lagrangian is

$$\begin{aligned} \mathcal{L}(\vec{x}, \vec{\lambda}) &= \frac{1}{2} \vec{x}^T Q \vec{x} + \vec{c}^T \vec{x} + \vec{\lambda}^T (A\vec{x} - \vec{b}) \\ &= \frac{1}{2} \vec{x}^T Q \vec{x} + (\vec{c} + A^T \vec{\lambda})^T \vec{x} - \vec{\lambda}^T \vec{b} \end{aligned}$$

$$\text{consider } \min \mathcal{L}(\vec{x}, \vec{\lambda}), \quad \frac{\partial \mathcal{L}(\vec{x}, \vec{\lambda})}{\partial \vec{x}} = Q\vec{x} + (\vec{c} + A^T \vec{\lambda}) = \vec{0}$$

$$\vec{x} = -Q^{-1}(\vec{c} + A^T \vec{\lambda})$$

Therefore substitute it back to $\mathcal{L}(\vec{x}, \vec{\lambda})$,

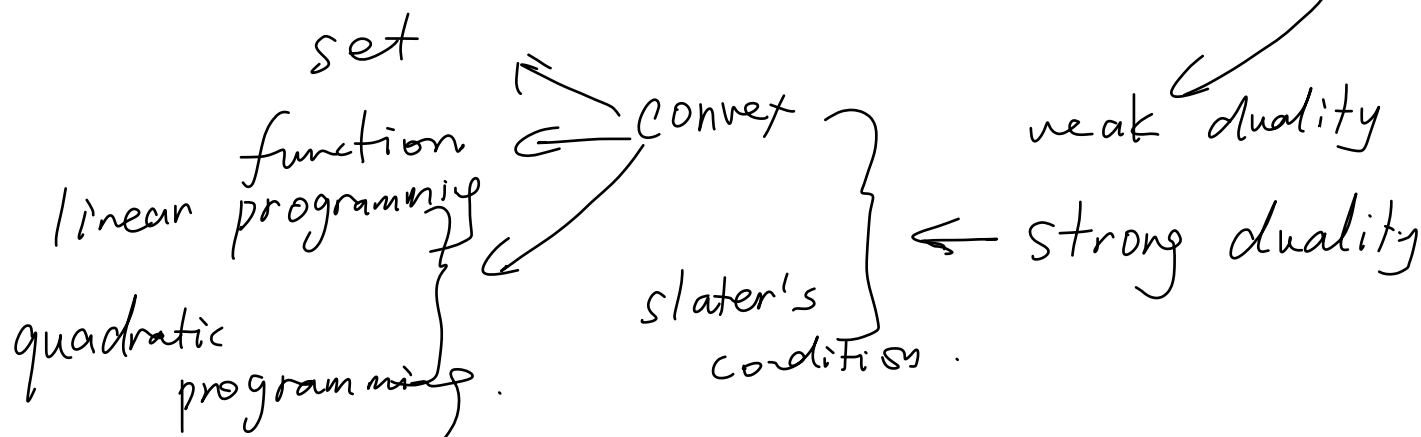
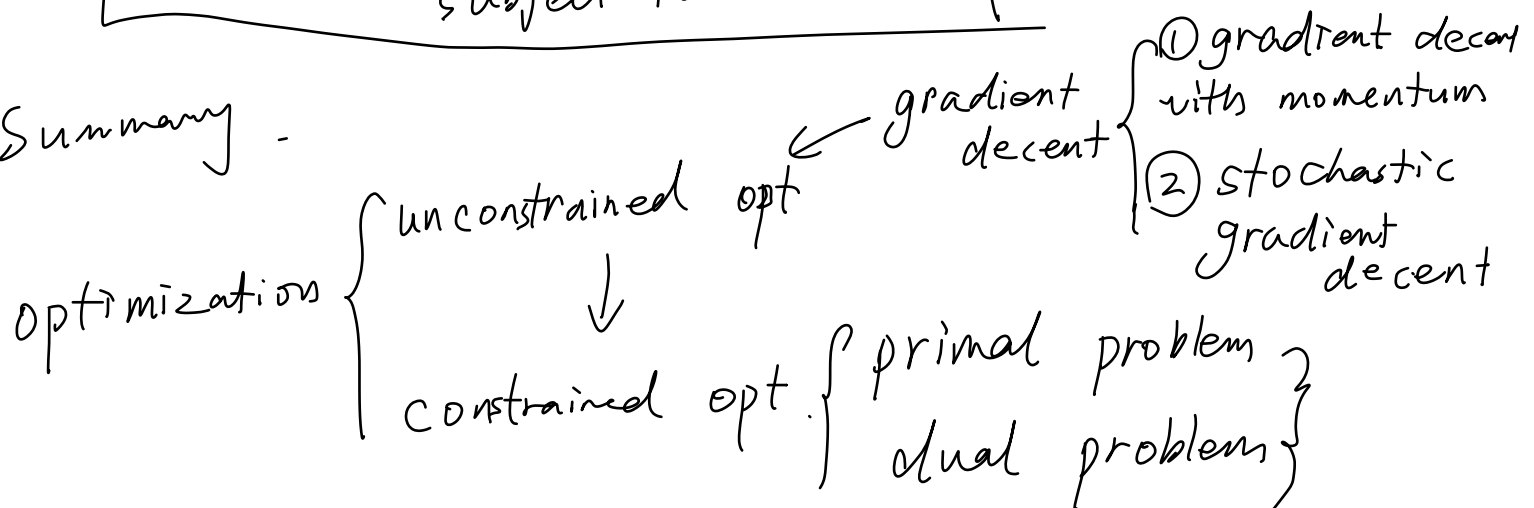
$$\Rightarrow D(\vec{\lambda}) = -\frac{1}{2} (\vec{c} + A^T \vec{\lambda})^T Q^{-1} (\vec{c} + A^T \vec{\lambda}) - \vec{\lambda}^T \vec{b}$$

dual optimization problem - ✓

$$\max_{\lambda \in \mathbb{R}^d} D(\vec{\lambda})$$

subject to $\vec{\lambda} \geq 0$

Summary -



section 7.1 ~ 7.3